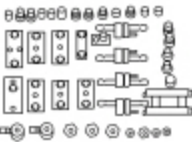


Air Conditioning (A/C) System Flushing



Special Tool(s)

 ST2985-A	A/C Flush Adapter Kit 219-00074 or equivalent
 ST2469-A	A/C Flush and Purge Fitting Kit 219-00024 or equivalent
 ST3313-A	A/C Flush and Purge Machine 357-00001 or equivalent

Material

Item	Specification
Motorcraft® A/C System Flushing Solvent YN-23	—
Motorcraft® PAG Refrigerant Compressor Oil YN-12-D	WSH-M1C231-B

WARNING: Take the following precautions when repairing an air conditioning system containing R-134a:

- Always wear safety goggles.
- Avoid contact with liquid refrigerant R-134a. R-134a vaporizes at approximately -25°C (-13°F) under atmospheric pressure and will freeze skin tissue.
- Never allow refrigerant R-134a gas to escape in quantity in an occupied space. It will displace the oxygen needed to support life.
- Never use a torch in an atmosphere containing R-134a gas. R-134a is non-toxic at all normal conditions, but it decomposes when exposed to high temperatures such as a torch flame. During decomposition it releases irritating and toxic gasses (as described in the Material Safety Data Sheet from the manufacturer). Decomposition products are hydrofluoric acid, carbon dioxide and water.

Failure to follow these instructions may result in serious personal injury.

NOTICE: An Air Conditioning (A/C) refrigerant analyzer must be used before the recovery of any vehicle's A/C refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes

to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. All contaminated A/C refrigerant must be disposed of as hazardous waste. For all equipment, follow the equipment manufacturer procedures and instructions.

NOTICE: Receiver/drier cartridge, front Thermostatic Expansion Valve (TXV) , rear TXV (if equipped) and hoses with mufflers should be removed when flushing the Air Conditioning (A/C) system. Internal plumbing of these devices makes it impossible to correctly remove any residual-flushing agent. These components are typically discarded after A/C system contamination. Hoses without mufflers can normally be reused unless they are clogged with foreign material. The 3.78L (1 gal) of Motorcraft® A/C System Flushing Solvent YN-23 and FL1-A filter used in A/C Flush and Purge Machine 357-00001 are intended for use on one vehicle only. They may be used to flush both the A/C condenser core and the A/C evaporator core on an individual vehicle, but under no circumstances should they be used on more than one vehicle.

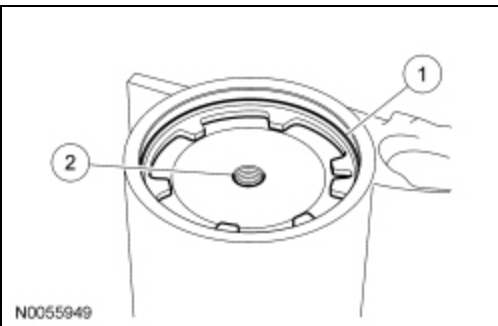
NOTICE: Only the A/C Flush and Purge Machine 357-00001 and Motorcraft® A/C Systems Flushing Solvent, is approved for use on Ford vehicles. No other flushing device or solvent is approved for flushing heat exchangers (Air Conditioning (A/C) condenser, A/C evaporator). Use of any other flusher or solvent may cause damage to the A/C system and the flushing unit.

NOTE: Prior to using the A/C Flush and Purge Machine 357-00001 for the first time, review the operating instructions.

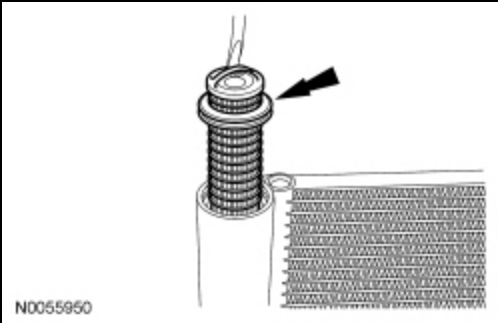
NOTE: Ford Motor Company has approved a procedure to provide technicians with a non-CFC method of flushing contaminated A/C system heat exchangers. The procedure allows the specific components to be cleaned and flushed. The types of contamination flushed include particle matter that results from A/C compressor or desiccant failure and gummy residue that can form when refrigerant oil is overheated during A/C compressor seizure. The flushing process is a 2-step procedure that involves the use of an A/C Flush and Purge Machine 357-00001 to:

- circulate the flushing solvent through the heat exchanger in the reverse direction of normal refrigerant flow (back-flushing). Particulate matter picked up during flushing is filtered from the returning solvent before the solvent is returned to the reservoir for continued circulation.
- remove the flushing solvent from the heat exchanger. In this step of the procedure, pressurized air 621-862 kPa (90-125 psi) is used to push and evaporate any remaining flush solvent from the heat exchanger.

1. Remove the condenser core. For additional information, refer to Section 412-01.
2. Remove the plastic receiver/drier cap.
3. Remove the receiver/drier plug.
 1. Push the receiver/drier plug towards the top of the condenser and remove the snap ring.
 2. Install an M5 bolt in the center of the receiver/drier plug and remove the plug.



4. Using a suitable tool, grasp the receiver/drier cartridge grab handle and remove the receiver/drier cartridge.



5. **NOTE:** *Do not install a new receiver/drier cartridge at this time.*

With the receiver/drier cartridge removed, reinstall the receiver/drier plug and snap ring into the condenser core.

6. Connect the A/C Flush and Purge Machine to the condenser inlet and outlet fittings using the correct adapters from the A/C Flush Adapter Kit.

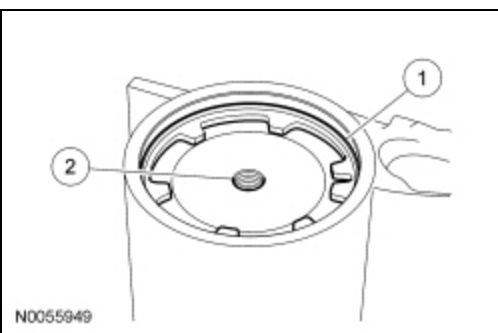
7. Flush the condenser core for a minimum of 15 minutes.

8. Turn the A/C Flush and Purge machine to the PURGE mode and run for a minimum of 30 minutes.

9. Disconnect the A/C Flush and Purge machine and adapters from the condenser core.

10. Remove the receiver/drier plug.

1. Push the receiver/drier plug towards the top of the condenser and remove the snap ring.
2. Install an M5 bolt in the center of the receiver/drier plug and remove the plug.



11. Install a new receiver/drier cartridge.

12. Install the new receiver/drier plug, snap ring and cap included in the receiver/drier cartridge service kit.

13. **NOTE:** *Do not charge the refrigerant system after the condenser core is installed.*

Install the condenser core. For additional information, refer to Section 412-01.

14. Remove the Thermostatic Expansion Valve (TXV) . For additional information, refer to Section 412-01.

15. Disconnect the evaporator inlet and outlet fittings.

- Discard the O-ring seals.

16. Connect the A/C Flush and Purge Machine to the evaporator inlet and outlet fittings using the correct adapters from the A/C Flush and Purge Fitting Kit.

17. Flush the evaporator core for a minimum of 15 minutes.

18. Turn the A/C Flush and Purge Machine to the PURGE mode and run for a minimum of 30 minutes.

19. Disconnect the A/C Flush and Purge Machine and adapters from the evaporator core.

20. Install new O-ring seals and connect the front evaporator inlet and outlet fittings.

21. Install a new front **TXV**, and new **TXV** gaskets. For additional information, refer to Section 412-01.

22. **NOTE:** *It is necessary to evacuate the refrigerant system for a minimum of one hour to evaporate any traces of flush solvent that may remain in the system.*

Evacuate the refrigerant system for one hour. For additional information, refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging in this section.

23. If a new A/C compressor is to be installed, install the new A/C compressor. For additional information, refer to Section 412-01.

24. If a new A/C compressor is not to be installed, add the correct amount of refrigerant oil to the A/C compressor in the following sequence.

1. Remove the A/C compressor. For additional information, refer to Section 412-01.
2. Drain the PAG oil and refill following the Refrigerant Oil Adding in this section.
3. Reinstall the A/C compressor. For additional information, refer to Section 412-01.

25. Evacuate, leak test and charge the refrigerant system. For additional information, refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging in this section.